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Powered Wing Response to Periodic Gust Encounters

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Propeller-wing interactions have been studied in great detail over several decades. However, almost all of them were conducted in steady initial conditions where the freestream remained constant. The objective of the current study is to explore the propeller-wing interactions in unsteady conditions and elucidate the role of thrust and power in a wing's response to a streamwise gust. An AR 2.5 Clark-Y full span wing model was integrated with propellers at different spanwise stations and was tested at different reduced frequencies. The periodic streamwise gusts were created in the University of Dayton Low Speed Wind Tunnel using the integrated shuttering system. The drag of the powered wing model at a low angle of attack was offset from the thrust using the propellers to simulate cruise conditions. Lift and pitching moment were measured and compared to a theoretical prediction using Isaacs' theory. The unsteady response of the wing to different propeller spanwise placement was quantified.

Nomenclature

AR	=	Aspect Ratio
C_L	=	Lift Coefficient, $C_L = \frac{L}{qS_{Ref}}$
C_M	=	Moment Coefficient, $C_M = \frac{M}{qS_{Ref}}$
L	=	Lift force, (N)
M	=	Pitching Moment, $(N.m)$
q	=	Dynamic Pressure, (Pa)
Re	=	Reynolds Number
S_{ref}	=	Reference area, (m^2)
V_∞	=	Freestream Velocity, $(\frac{m}{s})$
$\bar{V}$	=	Mean Velocity, $(\frac{m}{s})$
α	=	Angle of Attack, (Deg)
ρ	=	Air density, $(\frac{kg}{m^3})$
k	=	Reduced Frequency, $k = \frac{\omega c}{2\bar{V}}$
ω	=	Circular Frequency, (rad/s)
c	=	Chord Length, (m)
σ	=	Velocity Amplitude Ratio, $(\frac{V_{amp}}{\bar{V}})$
V_{amp}	=	Velocity Amplitude, (m/s)

I. Introduction

A century of research on wings, propellers, and their combination has led to the development and successful operation of commercial air vehicles and aircraft designs. However, the rise of battery technology and the need for clean and sustainable transportation has culminated in numerous Electric Vertical Takeoff and Landing (eVTOL) air vehicles called air taxis and Urban Air Mobility (UAM) vehicles. Drastic changes in design constraints must be made when compared to traditional aircraft, such as Vertical Takeoff and Landing (VTOL) capability, autonomy, small

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foot-print (compactness), energy efficiency, and a plethora of additional safety constraints when compared to conventional aircraft designs. Due to these changes, the air taxi designs stray far from the current norm leading to new research avenues on system integration (wing, structure, propulsion, control, etc.). Companies like Joby Aviation, Blade Air Mobility, and Volocopter are leading the charge in eVTOL air taxi development. These vehicles operate in low altitudes and are designed to takeoff and land on top of buildings, necessitating VTOL. In order to travel effectively, the vehicle must transition from vertical to forward flight. Historically, this transition is the most dangerous phase of flight, where unsteady conditions such as gusts can cause the aircraft to lose control. Building wakes and the atmospheric boundary layer can be highly turbulent and unsteady such that any vehicle operating in such conditions can lose control with potential for catastrophic failure.

Micro Air Vehicles (MAVs), specifically tail sitter configurations, have dealt with the transition between vertical and forward flight since their inception. Oftentimes, the transition is automatically stabilized by control systems set before flight due to the complexity of pure human-input control. Hostenbach et al. [1] found part of this complexity to be before the transition even begins – hover stability becomes an issue when the MAV encounters a wind gust. Once stabilized, their MAV will decrease pitch until the stall speed is exceeded and the optimal angle of attack is reached, at which point forward flight is achieved. A sudden loss or gain in altitude is common during the transition [2][3], which can be disastrous in the low altitude conditions air taxis operate under. Some controllers are able to counter the instability through mitigating the sharp changes in forces and moments produce by the MAV [2]. With control however, inaccurate gain scheduling can induce yaw oscillation when transitioning from forward to vertical flight [3]. While controllers are able to stabilize the transition, they can be flawed. A gust encounter during a controlled maneuver may bring even more instability to an already dangerous transition period.

A. Wing's Response to Streamwise Gusts

A wing's response to unsteady flow has been investigated through a variety of methods and gust profiles. Transverse, vortical, and streamwise gusts all have differing effects on the force response and the flow field. An overview of the three different types of gusts are discussed in Jones et al. [4]. The focus of the current study to determine the gust response of a standalone and a powered wing model and compare it to the theoretical predictions from Isaacs' theory [5]. Jones et al. [4] states that the change in lift coefficient through small velocity amplitude ratios is due to the local dynamic pressure change. In large velocity amplitude ratios, flow separation becomes the dominating factor in how lift changes. Strangfeld et al. [6] found that the theory developed by Isaacs can match experimental results with fairly respectable accuracy. In their tripped boundary layer condition, the theory accurately predicted lift trends and magnitudes. However, in lower Reynolds number regimes ($<10^6$) without a boundary layer trip, the separation bubble caused a lift coefficient undershoot along with high-frequency oscillations. Similar studies were conducted at a higher Reynolds number and discovered significant deviations in experimental results compared to the theory [7]. They found the Kutta condition to be violated for their streamwise gusts, with the stagnation line on the wing changing with the surging flow.

B. Propeller's Response to Streamwise Gusts

Similar to a wing, propellers encountering gusts have been investigated. Most research has been focused on the unsteady interactions between rotors and propeller blades [8][9]. Along with the development of air taxis, the focus has shifted to propellers interacting with an external source of unsteady flow. Cai et al. [10] found that steady-state performance between different velocities could be interpolated to develop a quasi-steady model of propellers in low reduced frequency (≤ 0.0796) streamwise gusts. Although some phase lag is introduced, the influence of the unsteady flow can be predicted, at least in the quasi-steady regime of flow. It was also found that propellers in streamwise forward flight have a different gust response when compared to edgewise flight. Others have found unsteady conditions to delay control inputs, affect power consumption, and overall performance [11][12].

C. Propeller-Wing Interactions

Propeller-wing systems are the prevailing design choice for air taxis and UAM vehicles due to their higher efficiency, lower weight, and lower cost when compared to jet engines under UAM operating conditions [13]. A powered wing will behave differently than an unpowered wing, due to propeller-wing interactions. A plethora of literature exists on active flow control through blown wings that are used to augment the lift coefficient by passing the engine wake on top of the wing, energizing the flow. Propellers will cause local upwash on the upward rotation, and a subsequent local downwash on its downward motion [14]. Energy transferred to the flow field from the propeller also increase the lift generated by the wing. Srivathsan and Rauleder [15] found that propellers delay separation on the upper surface of the wing, with an increased lift curve slope. The optimal downstream distance for the propeller to be located is one radius away from the wing, implying that optimal configurations for a propeller-wing system can be reached using a number of design considerations.

Almost no literature currently exists that explores the behavior of a powered wing to different types of gusts. The added thrust component, and the interactions between the propeller wake and the wing's boundary layer can cause drastic changes in the wing's response when compared to its standalone counterpart. As such, the aim of the current research is to characterize the role of thrust and propeller-wing interaction in wing's response to streamwise gusts.

II. Experimental Setup

All experiments were conducted in the University of Dayton Low Speed Wind Tunnel (UD-LSWT). The tunnel is an Eiffel type, open-jet configuration with six anti-turbulence screens, a 16:1 contraction ratio, and a 0.762m by 0.762m testing section. The testing section has a turbulence intensity of 0.1% measured by hot-wire anemometry, and is followed by a 1.370m by 1.370m collector. Placed directly after the collector is a state-of-the-art shuttering system, which has a set of rotating louvers. Changes in 1D flow velocity can be induced through systematic opening and closing of the louvers. The shuttering system was characterized to produce a sinusoidal freestream at a frequency range of 0Hz to 2.25Hz [10]. Figure 1 shows the open and closed configurations of the system, as well as the flow velocity change with different actuation frequencies at a wind tunnel fan rotational speed of 400 RPM. The flow velocity measurements were taken with hot-wire anemometry placed at the inlet of the testing section.

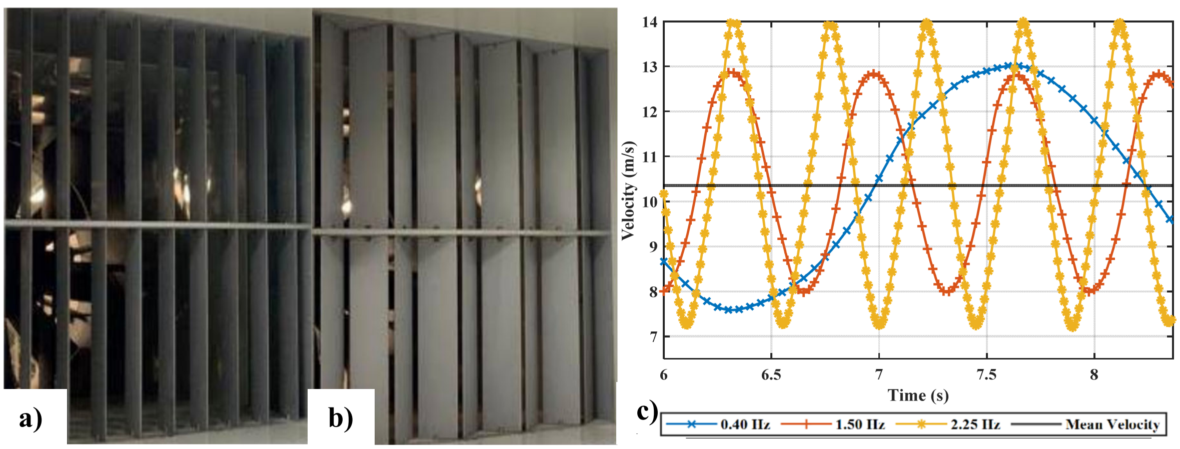

Figure 1. UD-LSWT Shuttering System a) Fully Open, b) Fully Closed, and c) Velocity Response to Actuation Frequency

The experimental setup is shown in Figure 2. The wing model has a Clark-Y airfoil with a chord of 0.203m and a span of 0.508m, corresponding to an aspect ratio of 2.5. The design includes three modular propeller locations (midspan, semi-midspan, and wingtip) for a total of 5 configurations. The two extra configurations arise from an unpowered version and two rotation directions for the wingtip configuration. The propulsion assembly consists of a Gemfan Hurricane 4024 2-blade, 0.102m diameter propeller and a T-Motor F2004 3000KV motor. The wing is mounted to an ATI Mini40-E force sensor [16], which is in turn mounted to a streamlined support system. A rotary stage on either side of the support allows the angle of attack of the entire system to be changed. Not pictured in Figure 2 is a Dantec Dynamics type 55R01 hot-wire [17] placed upstream at the inlet, used to collect instantaneous velocity measurements during actuation of the shuttering system.

The test matrix of the experiment is shown in Table 1, which includes both steady and unsteady conditions. The steady flow freestream velocity was set at 10 and 20 m/s, where the wing's angle of attack is varied from -6 to 30° in 2° increments. The unsteady flow mean velocity was 10.5 m/s, with a variation between 7 to 14 m/s. The angles of attack chosen were 0°, 5°, and 10°, with shuttering actuation frequencies of 0.4, 1.5, and 2.25 Hz. These shuttering actuation frequencies correspond to reduced frequencies (k) of 0.025, 0.101, and 0.136. The propellers were used to set a trimmed condition on the powered wing, meaning that at 0° angle of attack the propeller thrust matches the drag force of the propeller-wing system. The test matrix also includes wingtip configurations with opposite rotation.

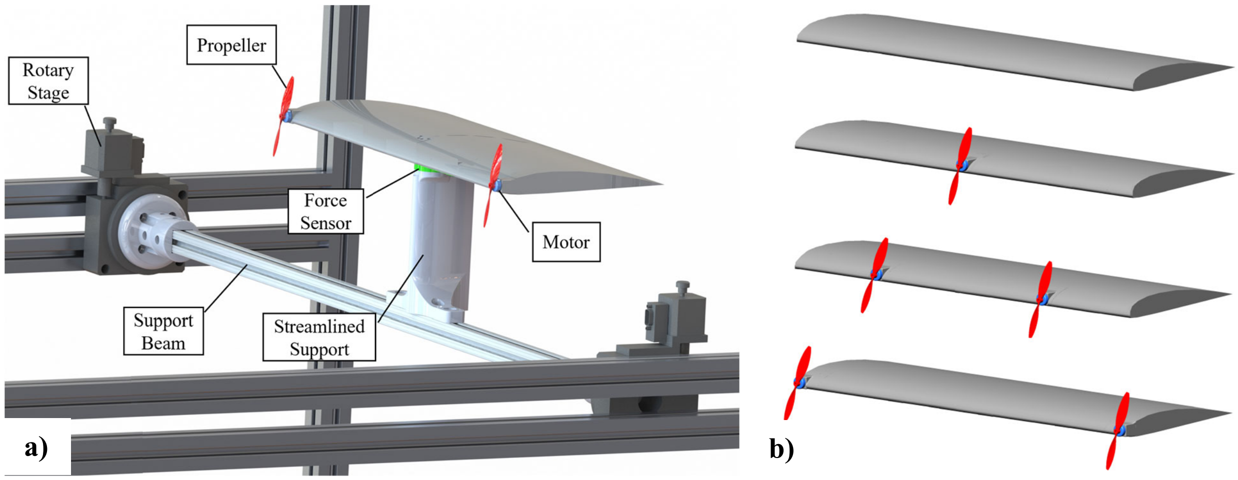

Figure 2. CAD Model for Experimental Setup of a) Wind Tunnel Testing and b) Propeller Locations

Table 1. Testing Matrix

Flow	Velocity (m/s)	Reynolds Number	Angle of Attack	Reduced Frequency (k) (Shuttering Frequency)	Configuration
Steady	10 20	130,000 260,000	-6° to 30°	N/A	Unpowered Midspan Semi-Midspan Wingtip: Inboard-Up Wingtip: Inboard-Down
Unsteady	10.5 mean	100,000 to 190,000	0°, 5°, 10°	0.025 (0.4 Hz) 0.101 (1.5 Hz) 0.135 (2.25 Hz)	

III. Isaacs's Theory

Many theoretical models have been developed to predict the lift response to unsteady flow, the most famous among them being Theodorsen [18], Wagner [19], and more. Each of these theories focus on different types of kinematics and unsteady flows, such as a pitching-plunging wing, a sharp-edged gust, sinusoidal gusts, vortex gusts, etc. Isaacs's theory [5] will be used for comparing the wing's response to sinusoidal gusts. Isaacs' theory assumes a 2D flat plate airfoil, where the wake travels downstream but always in the plane of the plate. The Kutta condition is always satisfied, constraining the theory to lower angles of attack where separation is limited or nonexistent. Velocity is simply composed of a constant part and a sinusoidally varying part, as seen in Equation 1.

$$u(t) = u_0(1 + \sigma \sin \omega t) \tag{1}$$

The inputs to Isaacs theory include σ, the velocity amplitude ratio, and k, the reduced frequency. Using a number of Bessel and Hankel functions, the output of Isaacs theory is the predicted lift divided by the steady state lift at the mean velocity u_0. The calculation of lift is below in Equation 2.

$$\frac{L}{L_0} = \frac{\sigma k}{2} \cos \omega t + \left(1 + \frac{\sigma^2}{2}\right)(1 + \sigma \sin \omega t) + \sigma \sum_{m=1}^{\infty} (l_m \cos m\omega t + l'_m \sin m\omega t) \tag{2}$$

Equation 2 is calculated using the following equations that are slightly modified from the Theodorsen function.

$$l_m + i l'_m = (-m)(-i)^m \sum_{n=1}^{\infty} \{F_n[J_{n+m}(n\sigma) - J_{n-m}(n\sigma)] + iG_n[J_{n+m}(n\sigma) - J_{n-m}(n\sigma)]\} \tag{3}$$

$$\left.\begin{matrix}F_n\\G_n\end{matrix}\right\} = \frac{J_{n+1}(n\sigma) - J_{n-1}(n\sigma)}{n^2}\left\{\begin{matrix}F(nk)\\G(nk)\end{matrix}\right. \tag{4}$$

$$C(nk) = F(nk) + iG(nk) = \frac{H_1^{(2)}(nk)}{H_1^{(2)}(nk) - iH_0^{(2)}(nk)} \tag{5}$$

van der Wall and Leishman [20] derived from another one of Isaacs's work [21] an equation for the pitching moment response through a streamwise gust (Equation 6), which is solved very similarly to Equation 2. $t_m + it'_m$ is solved the same as $l_m + il'_m$, with the only change coming from Equation 4. Equation 4 is replaced by Equation 7.

$$\frac{M}{M_0} = \left(1 + \frac{\sigma^2}{2}\right) + 2\sigma \sin \omega t - \frac{\sigma^2}{2} \cos 2\omega t + \sigma \sum_{m=1}^{\infty} (t_m \cos m\omega t + t'_m \sin m\omega t) \tag{6}$$

$$\left.\begin{matrix}F_n\\G_n\end{matrix}\right\} = \frac{J_{n+1}(n\sigma) - J_{n-1}(n\sigma)}{n^2}\left\{\begin{matrix}F(nk) - 1\\G(nk) - 1\end{matrix}\right. \tag{7}$$

The experimental results will be compared to Isaacs's theory for validation purposes. Although the experiment violates every one of Isaac's assumptions, based on [6][7] it is expected that his theory will predict similar magnitudes and trends.

IV. Results

A. Steady State Wing Response

The steady state coefficient of lift and pitching moment at different angles of attack will be discussed in this section. Figure 3a shows the variation of C_L with angle of attack for all of the configurations tested at 10 m/s. The baseline unpowered cased colored in black shows the expected linear variation in C_L with angle of attack until it stalled at an angle of attack of 18°. The effective AR of the wing using McCormick's formula was found to be 1.91. All the powered cases showed an increment in coefficient of lift at all angles of attack when compared to the baseline due to the added thrust vector. The addition of the thrust and the high velocity wake pushed the stall angle to higher angles of attack. The highest performer in terms of C_L magnitude is the midspan case, where the propeller is positioned at the wing root.

The difference between the powered and unpowered coefficient of lift performance is decreased at 20 m/s when compare to 10 m/s as seen in Figure 3b. This could be due to the increase in Reynolds number which also made the stall angle to be similar in all cases. Therefore, when compared to an unpowered wing, the powered wing performance in C_L is only marginally higher. However, significantly higher lift is seen in all the post stall regions in the powered wing cases when compared to the unpowered case.

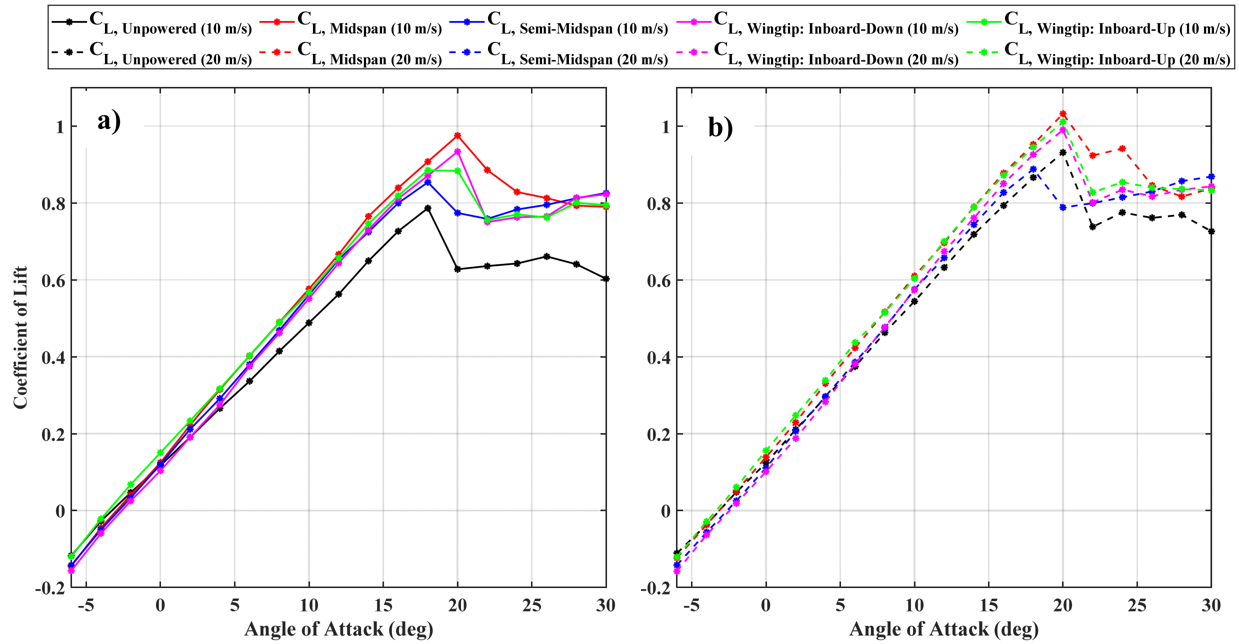

Figure 3. C_L vs. Alpha for Steady Flow

Figure 4 displays the coefficient of pitching moment between all configurations, with Figure 4a containing the results at 10 m/s and Figure 4b containing the results at 20 m/s. When compared to the unpowered model, the addition of propeller thrust increased the magnitude of the moment coefficient, due to the thrust vector being applied above the force sensor. This effect decreases as the propeller moves towards the wingtips. Wake velocity and wing area coverage play key factors in the moment coefficient. The Midspan and Wingtip configurations have the same total wing area coverage of 20%. However, less lift is generated at the wingtips, leading to a smaller influence. The Midspan configuration also has a single propeller, which operated at a higher rotational speed when compared to the two propeller configurations in order to achieve cruise trim condition. Once again, the direction of propeller rotation at the wingtips made insignificant difference in both C_L and C_M, indicating that there is no added benefit or detriment to propeller rotation direction with or against the direction of the wingtip vortex rollup.

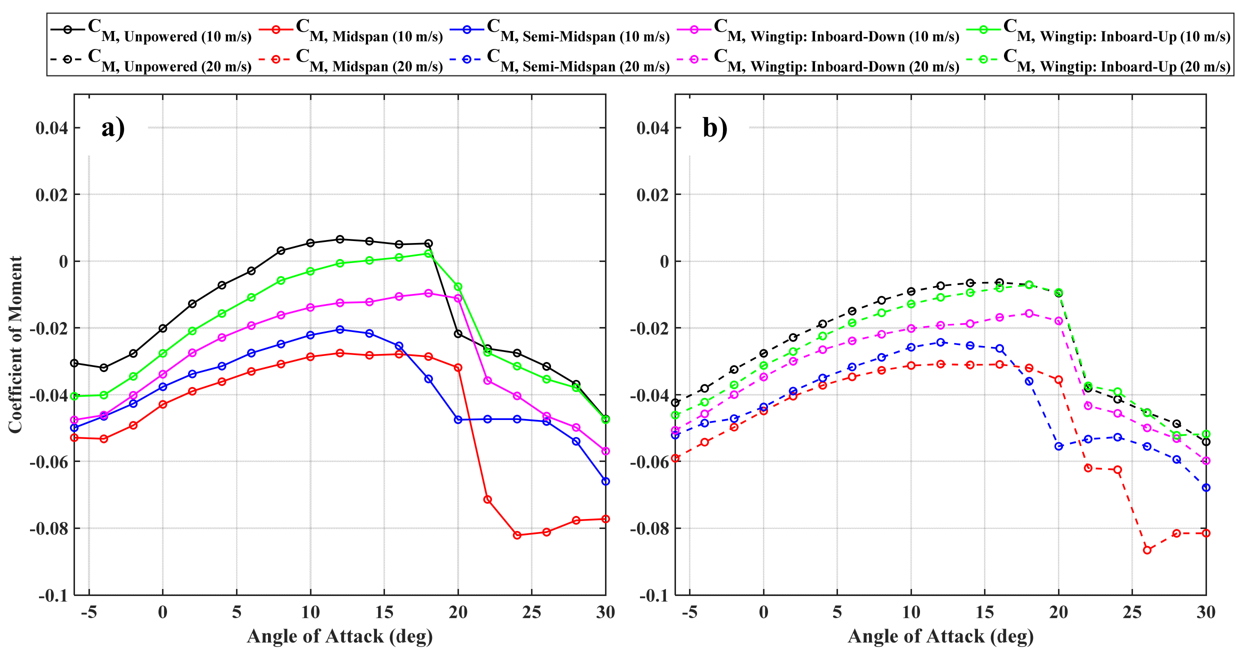

Figure 4. C_M vs. Alpha for Steady Flow

B. Wing's Response to Periodic Streamwise Gusts

The wing's unsteady response to a periodic streamwise gust is shown in Figure 5 when the wing is placed at 0° angle of attack. The first column shows the C_L vs. time at different reduced frequencies and the second column shows the C_M vs. time at different reduced frequencies. The plots also compare the unpowered and midspan configurations to Isaacs' theory. The y-axis on the left of each plot corresponds to the magnitude of C_L, while the y-axis on the right corresponds to the change in velocity from the mean of 10.5 m/s. The fluctuation in C_L and C_M increase with reduced frequency. Lift coefficient reaches its local maximum when the freestream velocity reaches its local minimum. However as pitching moment coefficient reaches its local maximum, the freestream velocity reaches its local maximum as well. For a reduced frequency of 0.136, a sudden change in both C_L and C_M occurs at the trough of the freestream velocity. At the velocity peak, a gradual response is shown from both coefficients. Isaacs' theory deviates from the experimental results in both phase and magnitude in the lowest reduced frequency, but more closely represents the response at the higher reduced frequencies. However, the predictions for C_M show very small perturbations even at the higher reduced frequency, and cannot replicate most of the trend shown in the time history.

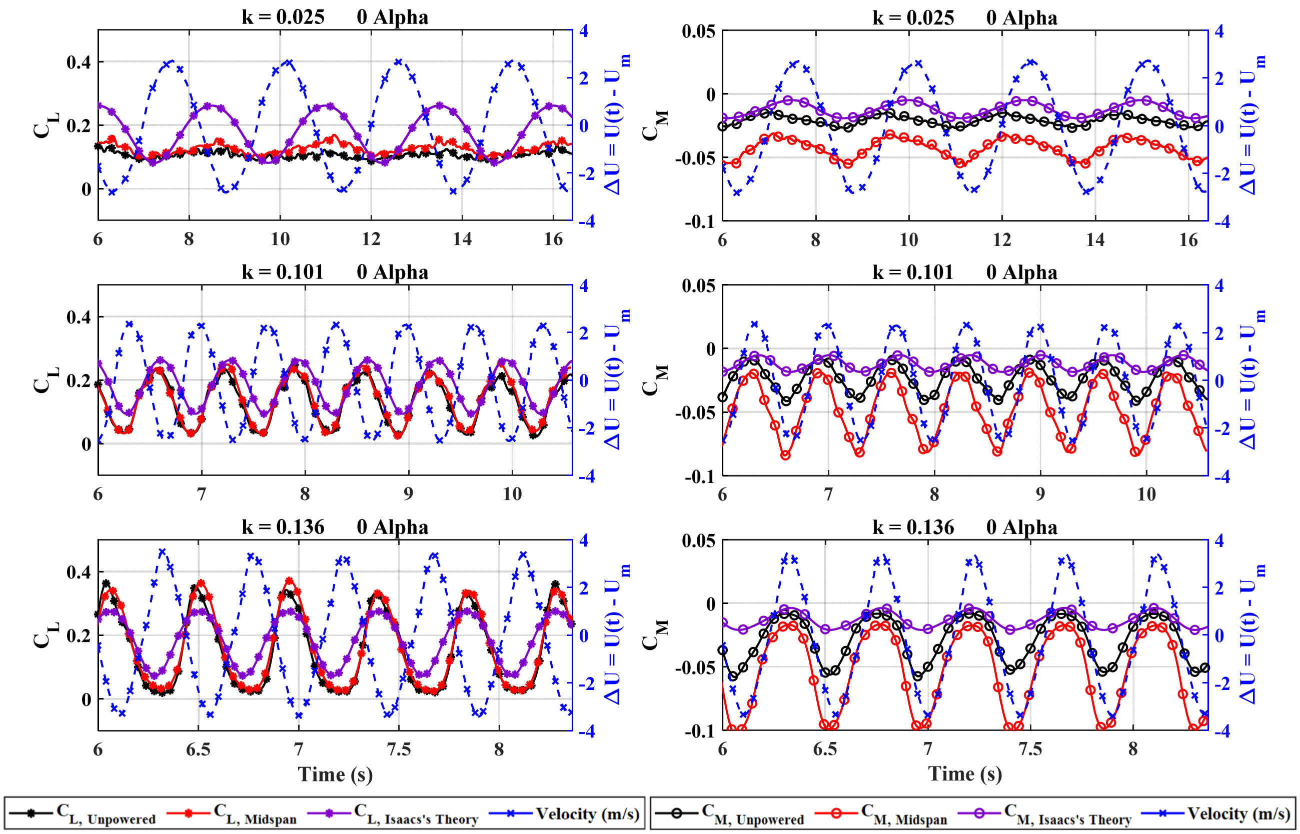

Figure 5. C_L, C_M vs. Time for Unpowered and Midspan Configurations Compared to Isaacs's Theory

Figure 6 replots the C_L and C_M data shown in Figure 5 against the change in velocity instead of time. In this case, the trends between configurations, angles of attack, and reduced frequency can be studied with ease. The direction of velocity change is counter-clockwise, meaning that as C_L decreases the velocity increases, and vice versa. This "counter-clockwise rotation" can be thought of as the time direction in this visualization. The first row consists of the quasi-steady responses at 0°, 5°, and 10°, where a small change in magnitude is expected and matches with Figure 5. With an increase in angle of attack, the fluctuation increases and a hysteresis loop forms. Higher phase lag manifests in the form of a larger hysteresis loop and vice versa. The wing will have a different response in lift for an increasing versus a decreasing velocity. Recall from Figure 5, a sharp and gradual response of the wing performance is observed at $k = 0.136$, which is also shown in the hysteresis where a sharp response of C_L occurs at peak velocity, which differs from lower reduced frequency cases. With velocity at its peak, the C_L has relatively similar values which leads to the decrease in phase lag and relates to the gradual response above. With velocity at its trough, the C_L has a range of values leading to an increase in phase lag and relates to the sharp response above. Overall, the same trends can be seen between the Unpowered and the Midspan configurations. The powered configuration has a higher magnitude, which agrees with the steady state data from Figure 3.

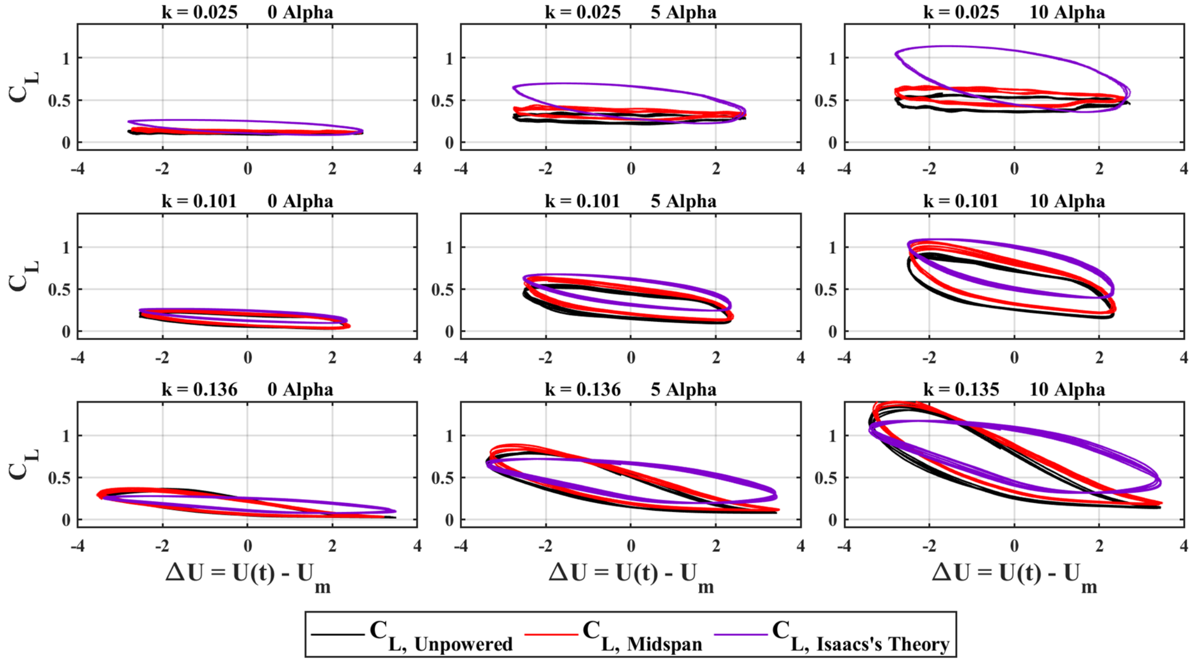

Figure 6. C_L vs. ΔU for Unpowered and Midspan Configurations Compared to Isaacs's Theory

Figure 7 compares the performance of all the powered cases against each other at different angles of attack and reduced frequencies. Similar to Figure 6, the C_L magnitude fluctuation and phase lag increase with an increasing angle of attack and reduced frequency. The trend is skewed to show an increasing velocity corresponds to a decrease in C_L, and vice versa. In general, each of the powered plots overlap on each other, meaning that their trends and magnitudes are all the same. The only difference seen in this plot is once again the magnitude increase with the powered configurations compared to the unpowered one.

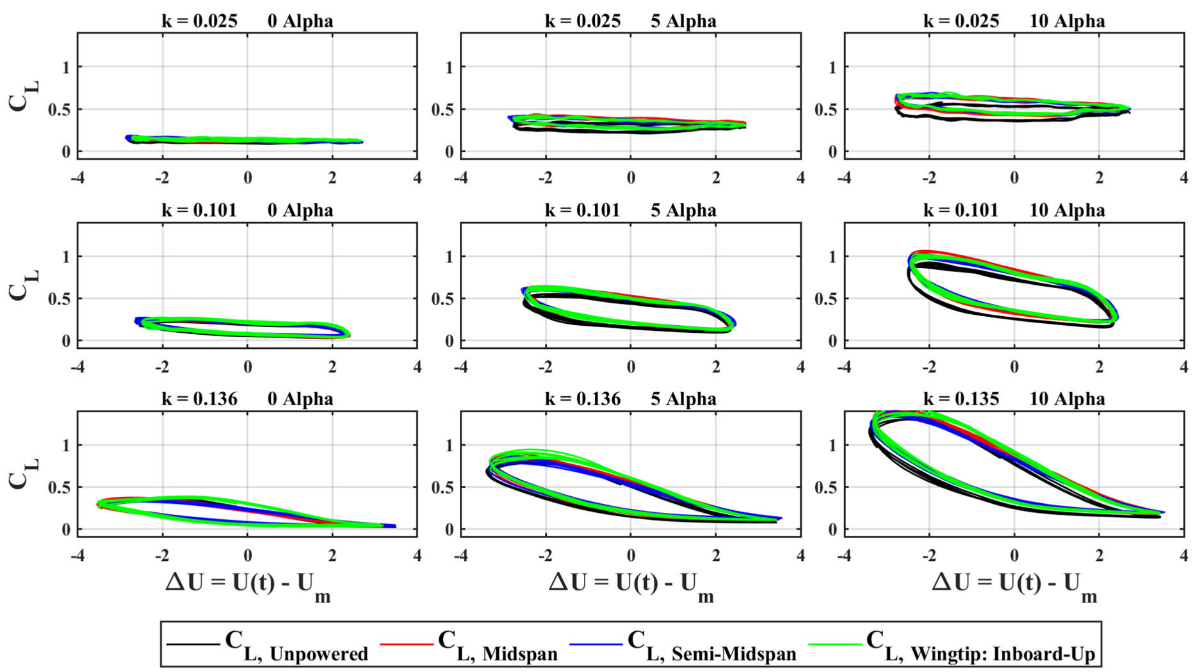

Figure 7. C_L vs. ΔU for All Configurations

Figure 8 contains the same comparison as Figure 6 for pitching moment coefficient. An obvious magnitude difference can be seen between cases, which is replicated in the steady state data seen in Figure 4. The direction of velocity is once again counter-clockwise, meaning that as velocity increases the moment coefficient does as well. The hysteresis loop is increased for the Midspan case across all angles of attack and reduced frequencies. The trends between wings remain very similar throughout all conditions as well. Isaacs' theory struggles to replicate any of the phase lag except for the lowest reduced frequency, where the lowest phase lag is also expected. It cannot replicate the trends seen in C_M nearly as well as it could for C_L.

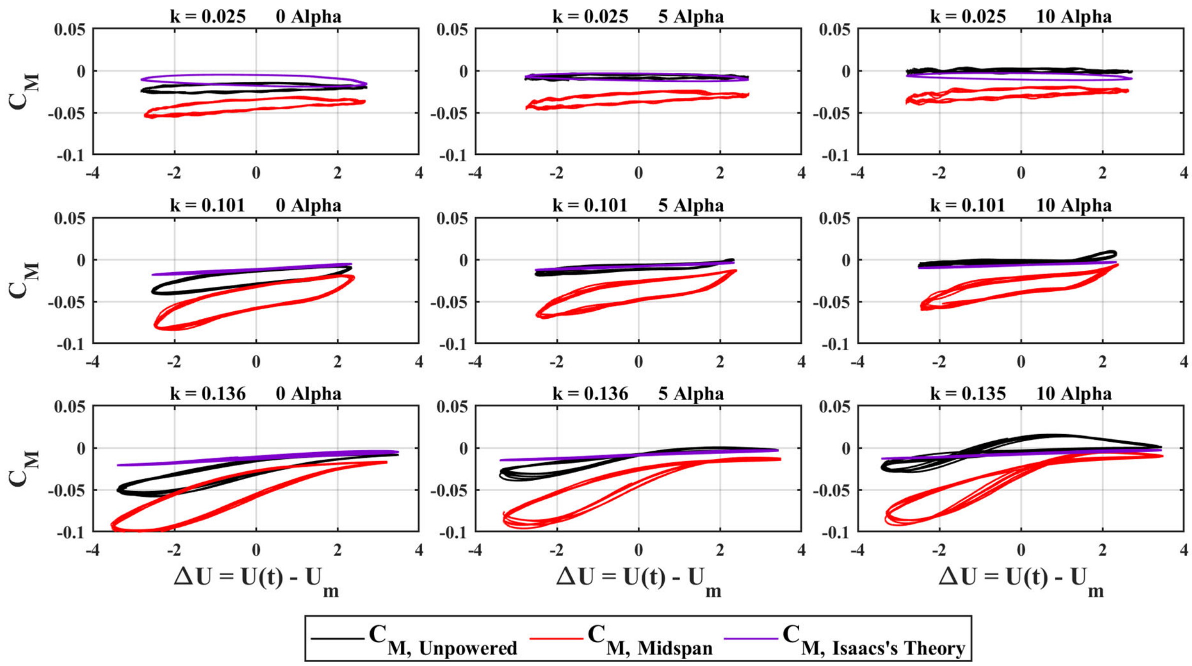

Figure 8. C_M vs. ΔU for Unpowered and Midspan Configurations Compared to Isaacs's Theory

Figure 9 shows the C_M results for all other models. The trends originally seen in the Unpowered and Midspan comparison are replicated in the other two powered configurations. Again, the only difference is in magnitude of C_M, which is seen in the steady state data (Figure 4) and therefore is not an effect of the unsteady interactions.

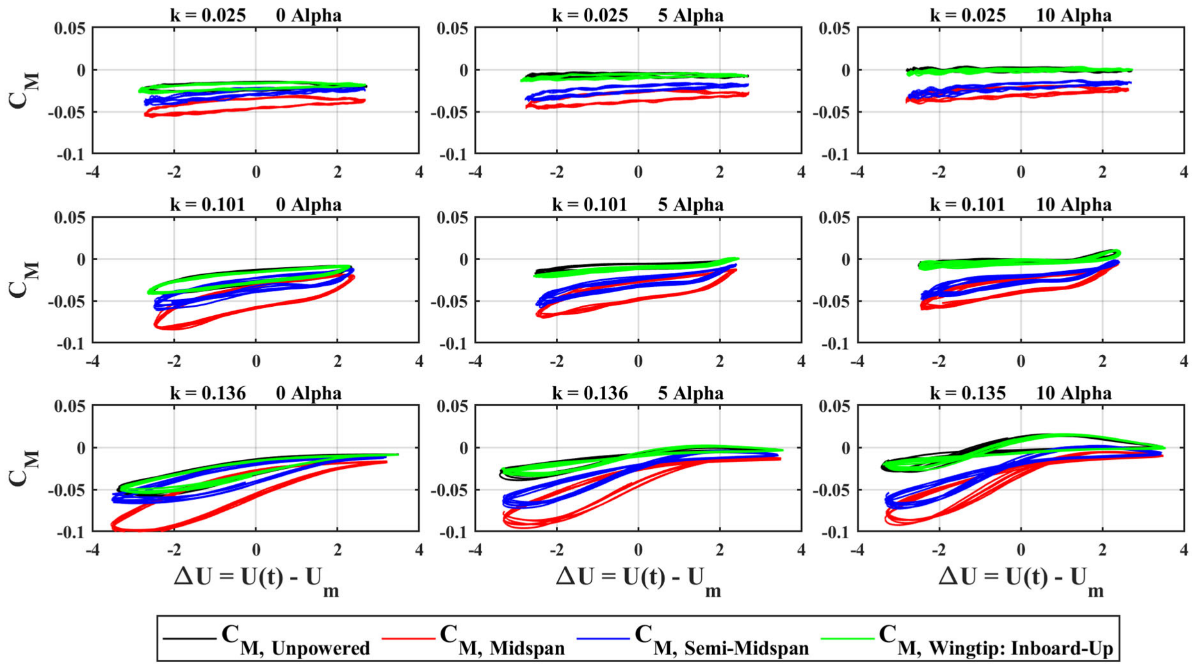

Figure 9. C_M **vs. ΔU for All Configurations**

V. Conclusion

A powered wing's response to a streamwise gust deviates slightly from the response of an unpowered wing. This could be due to the fact that the increment in lift in steady state is also marginal and were not significant beyond stall. Although the trends through a streamwise gust are very similar between cases, he coefficients of lift and moment increase in magnitude with the addition of a propeller. The propeller wake velocity and wing area coverage are the most likely contributing factors to the increment in lift and moment under unsteady conditions, as the section covered by the wake turns into a blown wing configuration. Isaac's theory closely predicts the experimental response, especially at low angles of attack where separation is least likely to occur. The deviations from experimental results can be attributed to the assumptions Isaacs made when developing his theory.

VI. Future Work

Future work will involve a full spatial characterization of the unsteady flow during the actuation of the shuttering system at different frequencies. The spatial velocity information will be used to determine the coefficients more accurately than a single point velocity measurement. The thrust from the propellers will also be increased to provide maximum contribution of thrust towards the lift coefficient, and a similar study will be conducted. The effects of camber will be studied by comparing the results to a symmetrical wing. The response to discrete streamwise gusts must also be captured in the form of step and impulse velocity changes in order to build a closed loop controller designed to ensure safe travel through these gusts.

Acknowledgments

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